

Question 1 **[Maximum mark: 10]** ***[Start on a new green sheet]***

(a) Evaluate $\sqrt{\frac{\pi^2 + 3.2^3}{5.6 - 1.2^2}}$ correct to **three** significant figures. **2**

(b) Simplify $\frac{x}{x^2 - 4} + \frac{2}{x - 2}$ **2**

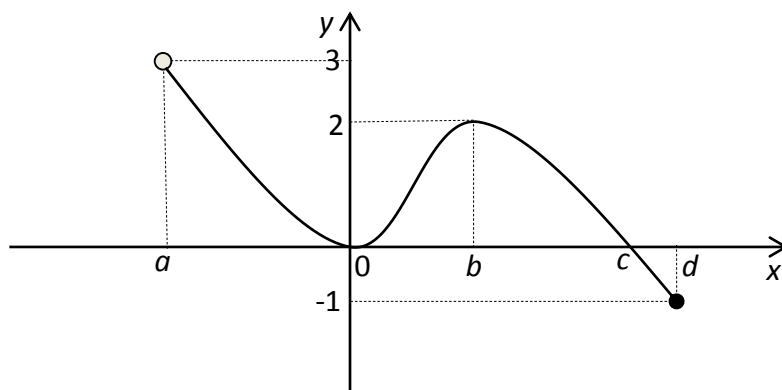
(c) Find integers a and b such that $\frac{1}{\sqrt{5} - 2} = a + b\sqrt{5}$ **2**

(d) Show that $f(x) = x^4 + 3x^2 - 1$ is an even function. **2**

(e) Solve the inequality $x^2 - x - 12 < 0$ **2**

Question 2 **[Maximum mark: 10]** **[Start on a new green sheet]**

- (a) The diagram shows a sketch of the curve $y = g(x)$ 3



- (i) Write down the range of $g(x)$.
- (ii) For what values of x is $g(x)$
- Increasing
 - Decreasing
- (b) Shade the region in the Cartesian plane for which the following inequalities hold simultaneously.

$$y < x - 2, \quad y \geq 0 \quad \text{and} \quad x \geq 6 \quad \quad \quad 3$$

(c) Simplify : $\frac{6^m + 2^m}{2^m}$ 2

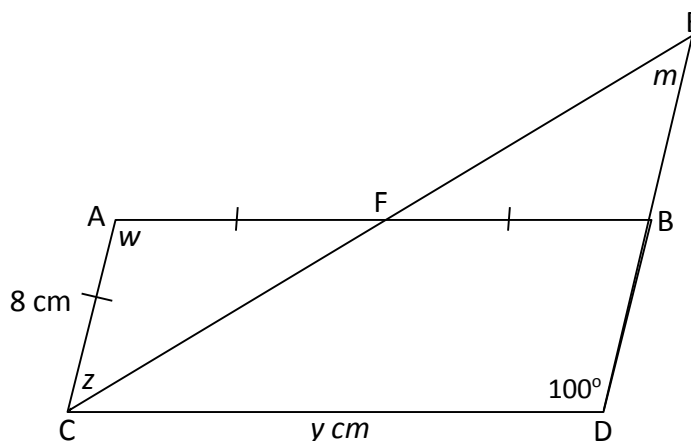
(d) Solve $9^{x-3} = 3$ 2

Question 3 [Maximum mark : 10]**[Start on a new green sheet]**

(a) Evaluate the following limits:

$$(i) \quad \lim_{x \rightarrow 2} \frac{2x+3}{x-1} \quad 1$$

$$(ii) \quad \lim_{x \rightarrow 1} \frac{x^3-1}{x-1} \quad 2$$

(b) ABCD is a parallelogram. Find the values of m, w, y and z. 4

(c) Sketch the following graphs and state their domain.

$$(i) \quad y = \sqrt{16 - x^2} \quad 1$$

$$(ii) \quad y = \frac{3}{x-2}, \quad -3 \leq y \leq -1 \quad 2$$

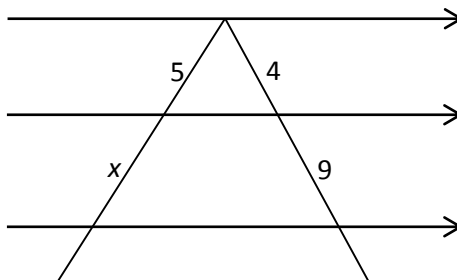
Question 4

[Maximum mark: 10]

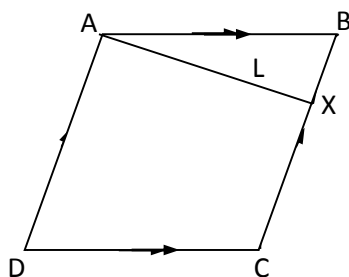
[Start on a new green sheet]

- (a) Find the value of
- x
- .

2



- (b)
- $ABCD$
- is a rhombus,
- AX
- is perpendicular to
- BC
- and intersects
- BD
- at
- L
- .



Not to scale

- (i) Copy the diagram into your Answer sheet and explain why

$$\angle ADB = \angle CDB.$$

2

- (ii) Prove that
- $\triangle ALD$
- and
- $\triangle CLD$
- are congruent.

3

- (iii) Show that
- $\angle DAL$
- is a right angle.

2

- (iv) Hence or otherwise find the size of
- $\angle LCD$

1

End of Examination



✓ = 1/2 mark
✓ = 1 mark

Student Number:

QUESTION 1.

(a) 3.20 ✓ (2) [3 20147218...]

(e) $x^2 - x - 12 < 0$

$(x-4)(x+3)$ ✓

$x = 4, -3$

(b) $\frac{x}{x^2-4} + \frac{2}{x-2}$

$\frac{1}{-3} \quad \frac{1}{4}$ ✓

$\frac{x+2(x+2)}{(x+2)(x-2)}$ ✓

$\therefore -3 < x < 4$

$= \frac{x+2x+4}{(x+2)(x-2)}$ ✓

(2)

$= \frac{3x+4}{(x+2)(x-2)}$ ✓ (2)

(c) $\frac{1}{\sqrt{5}-2} \times \frac{\sqrt{5}+2}{\sqrt{5}+2}$ ✓

$= \frac{\sqrt{5}+2}{(\sqrt{5})^2-2^2} = \frac{\sqrt{5}+2}{5-4}$



$= \sqrt{5}+2$ ✓

$\therefore a = 2$ ✓ (2)

$b = 1$ ✓

(d) $f(x) = x^4 + 3x^2 - 1$

$f(-x) = (-x)^4 + 3(-x)^2 - 1$ ✓

$= x^4 + 3x^2 - 1$ ✓

$= f(x)$ ✓

(2)

$\therefore f(x)$ is even.



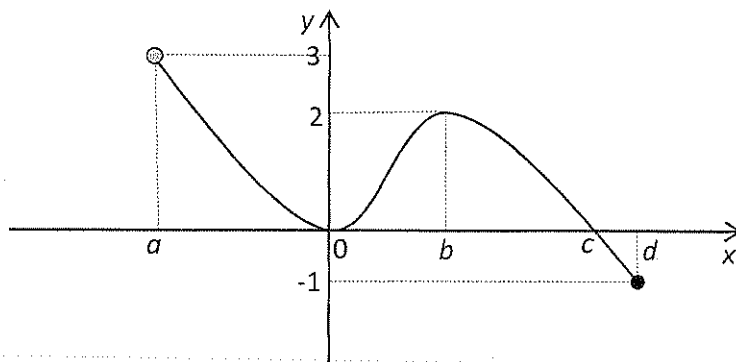
Question 2

[Maximum mark: 10]

[Start on a new green sheet]

- (a) The diagram shows a sketch of the curve
- $y = g(x)$

3



$\frac{1}{2}$ mark in total
taken for
 $\leq$ instead of $<$

- (i) Write down the range of
- $g(x)$
- .

$$\{y: -1 \leq y < 3, y \in \mathbb{R}\}$$

- (ii) For what values of
- x
- is
- $g(x)$

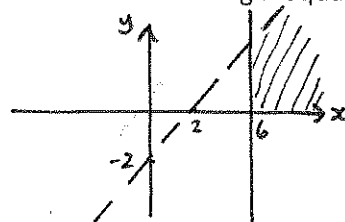
- Increasing
- Decreasing

$$0 < x < b$$

$$a < x < 0, b < x < d$$

- (b) Shade the region in the Cartesian plane for which the following inequalities hold simultaneously.

$$y < x - 2, y \geq 0 \text{ and } x \geq 6$$



- Diagram should be $\frac{1}{3}$ to $\frac{1}{2}$ page.
- Ruler should be 3 used
- At least one number should be used on each axis to show scale.

(c) Simplify: $\frac{6^m + 2^m}{2^m} = \frac{3^m 2^m + 2^m}{2^m} = \frac{2^m (3^m + 1)}{2^m} = 3^m + 1$

Reduce to primes

- (d) Solve
- $9^{x-3} = 3$

$$3^{2(x-3)} = 3^1 \quad \checkmark$$

$$\therefore 2(x-3) = 1 \quad \checkmark$$

$$x-3 = \frac{1}{2}$$

$$x = \frac{1}{2} + 3$$

$$x = 3\frac{1}{2} \text{ or } \frac{7}{2} \quad \checkmark$$

OR

$$9^{x-3} = 9^{\frac{1}{2}}$$

$$\therefore x-3 = \frac{1}{2}$$

$$x = 3\frac{1}{2}$$

OR

$$\log_9 9^{x-3} = \log_9 3$$

$$0.5 = x-3$$

$$3.5 = x$$

OR

$$\log_{10} 9^{x-3} = \log_{10} 3$$

$$(x-3) \log_{10} 9 = \log_{10} 3$$

$$x-3 = \frac{\log_{10} 3}{\log_{10} 9}$$

$$x-3 = \frac{1}{2}$$

$$x = 3\frac{1}{2}$$



$x = \frac{1}{2}$ mark
 $\checkmark = 1$ mark

Student Number:

QUESTION 3.

(a) (i)

$$\lim_{x \rightarrow 2} \frac{2x+3}{x-1}$$

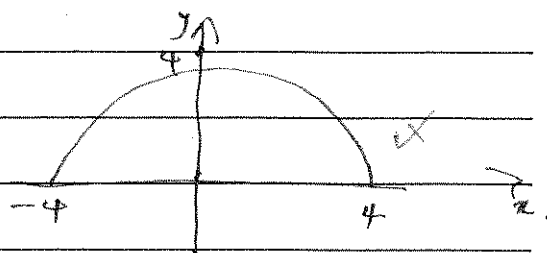
$$= \frac{2(2)+3}{2-1}$$

$$= \underline{7} \quad \checkmark \quad (1)$$

(c) (i)

$$y = \sqrt{16-x^2}$$

$$= \sqrt{4^2-x^2}$$



(v) $\lim_{x \rightarrow 1} \frac{x^3-1}{x-1}$

$$= \lim_{x \rightarrow 1} \frac{(x-1)(x^2+x+1)}{x-1} \quad \checkmark$$

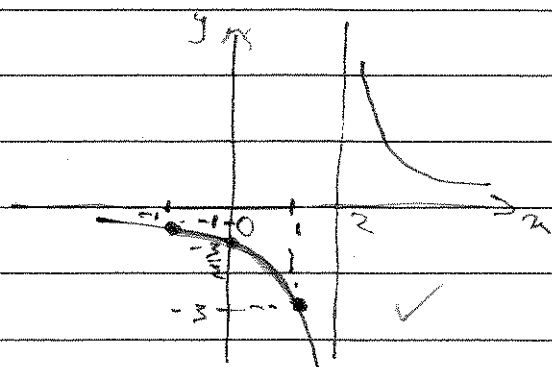
$$= \lim_{x \rightarrow 1} x^2+x+1$$

$$= 1^2+1+1 \quad \checkmark$$

$$= \underline{3} \quad \checkmark \quad (2)$$

Domain: $-4 \leq x \leq 4$. $\checkmark$ (1)

(ii) $y = \frac{3}{x-2}$



(b) $M = 40^\circ \quad \checkmark$

$W = 100^\circ \quad \checkmark$

$y = 16 \text{ cm} \quad \checkmark$

$z = 40^\circ \quad \checkmark$

(4)

$$-3 = \frac{3}{x-2}$$

$$|-1 = \frac{3}{x-2}$$

$$-3(x-2) = 3$$

$$-1(x-2) = 3$$

$$-3x+6 = 3$$

$$-x+2 = 3$$

$$-3x = -3$$

$$-x = 1$$

$$x = 1$$

$$x = -1$$

Domain: $-1 \leq x \leq 1$ $\checkmark$

(2)



$x = \frac{1}{2}$ mark
 $\checkmark = 1$ mark

Student Number:

QUESTION 4.

(a) $\frac{x}{5} = \frac{9}{4}$ ✓

$4x = 45$ ✓

$x = \frac{45}{4} = 11\frac{1}{4}$ ✓

(2)

(b) (i) The diagonals of a rhombus bisect the angles of a rhombus.

1 mark for the diagram

1 mark for the reason.

(2)

(ii) In $\triangle ACD$ and $\triangle CLD$

DL is common ✓

✓ $AD = DC$ Sides of a rhombus ✓

$\angle ADL = \angle CDL$ (i) above. ✓

$\therefore \triangle ACD \equiv \triangle CLD$ (SAS) ✓ (3)

(iii) $\angle DAL = 90^\circ$ (Alt. angles on // lines). ✓ (2)

(iv) $\angle LCD = 90^\circ$ corresponding angles of congruent $\triangle$. ✓ (1)